

Rewriting the marriage condition in this way does indeed yield an infinite version of Hall's theorem, which follows from Theorem 8.4.10 just as the marriage theorem follows from König's theorem:

Corollary 8.4.11. *A bipartite graph with bipartition $\{A, B\}$ contains a matching of A unless there is a set $S \subseteq A$ such that S is not matchable to $N(S)$ but $N(S)$ is matchable to S .*

Proof. Consider a matching M and a cover U as in Theorem 8.4.10. Then $U \cap B \supseteq N(A \setminus U)$ is matchable to $A \setminus U$, by the edges of M . And if $A \setminus U$ is matchable to $N(A \setminus U)$, then adding this matching to the edges of M incident with $A \cap U$ yields a matching of A . $\square$

Applied to a finite graph, Corollary 8.4.11 implies the marriage theorem: if $N(S)$ is matchable to S but not conversely, then clearly $|S| > |N(S)|$. Similarly, the finite version of Corollary 8.4.11 implies the finite case of the following sufficient condition for the existence of a matching of A :

Theorem 8.4.12. (Milner & Shelah 1974)

A bipartite graph with bipartition $\{A, B\}$ contains a matching of A if $d(a) \geq 1$ for every $a \in A$ and $d(a) \geq d(b)$ for every edge ab with $a \in A$.

*partial
matching*

*augmenting
path*

Let us now turn to non-bipartite graphs. If a finite graph has a 1-factor, then the set of vertices covered by any *partial matching* – one that leaves some vertices unmatched – can be increased by an augmenting path, an alternating path whose first and last vertex are unmatched (Ex. 1, Ch. 2). In an infinite graph we no longer insist that augmenting paths be finite, as long as they have a first vertex. Then, starting at any unmatched vertex with an edge of the 1-factor that we are assuming to exist, we can likewise find a unique maximal alternating path that will either be a ray or end at another unmatched vertex. Switching edges along this path we can then improve our current matching to increase the set of matched vertices, just as in a finite graph.

The existence of an inaugmentable partial matching, therefore, is an obvious obstruction to the existence of a 1-factor. The following theorem asserts that this obstruction is the only one:

Theorem 8.4.13. (Steffens 1977)

A countable graph has a 1-factor if and only if for every partial matching there exists an augmenting path.

Unlike its finite counterpart, Theorem 8.4.13 is far from trivial: augmenting a given matching ‘blindly’ need not lead to a well-defined matching at limit steps, since a given edge may get toggled on and off infinitely often (in which case its status will be undefined at the limit – example?). We therefore cannot simply find the desired 1-factor inductively.

In fact, Theorem 8.4.13 does not extend to uncountable graphs (Exercise 73). However, from the obstruction of inaugmentable partial matchings one can derive a Tutte-type condition that does extend. Given a set S of vertices in a graph G , let us write $\mathcal{C}'_{G-S}$ for the set of factor-critical components of $G - S$, and G'_S for the bipartite graph with vertex set $S \cup \mathcal{C}'_{G-S}$ and edge set $\{sC \mid \exists c \in C: sc \in E(G)\}$.

$\mathcal{C}'_{G-S}$
 G'_S

Theorem 8.4.14. (Aharoni 1988)

A graph G has a 1-factor if and only if, for every set $S \subseteq V(G)$, the set $\mathcal{C}'_{G-S}$ is matchable to S in G'_S .

Applied to a finite graph, Theorem 8.4.14 implies Tutte's 1-factor theorem (2.2.1): if $\mathcal{C}'_{G-S}$ is not matchable to S in G'_S , then by the marriage theorem there is a subset S' of S that sends edges to more than $|S'|$ components in $\mathcal{C}'_{G-S}$ that are also components of $G - S'$, and these components are odd because they are factor-critical.

Theorems 8.4.10 and 8.4.14 also imply an infinite version of the Gallai-Edmonds theorem (2.2.3):

Corollary 8.4.15. *Every graph $G = (V, E)$ has a set S of vertices that is matchable to $\mathcal{C}'_{G-S}$ in G'_S and such that every component of $G - S$ not in $\mathcal{C}'_{G-S}$ has a 1-factor. Given any such set S , the graph G has a 1-factor if and only if $\mathcal{C}'_{G-S}$ is matchable to S in G'_S .*

Proof. Given a pair (S, M) where $S \subseteq V$ and M is a matching of S in G'_S , and given another such pair (S', M') , write $(S, M) \leq (S', M')$ if

$$S \subseteq S' \subseteq V \setminus \bigcup \{V(C) \mid C \in \mathcal{C}'_{G-S}\}$$

and $M \subseteq M'$. Since $\mathcal{C}'_{G-S} \subseteq \mathcal{C}'_{G-S'}$ for any such S and S' , Zorn's lemma implies that there is a maximal such pair (S, M) .

S, M

For the first statement, we have to show that every component C of $G - S$ that is not in $\mathcal{C}'_{G-S}$ has a 1-factor. If it does not, then by Theorem 8.4.14 there is a set $T \subseteq V(C)$ such that $\mathcal{C}'_{C-T}$ is not matchable to T in C'_T . By Corollary 8.4.11, this means that $\mathcal{C}'_{C-T}$ has a subset $\mathcal{C}$ that is not matchable in C'_T to the set $T' \subseteq T$ of its neighbours, while T' is matchable to $\mathcal{C}$; let M' be such a matching. Then $(S, M) < (S \cup T', M \cup M')$, contradicting the maximality of (S, M) .

Of the second statement, only the backward implication is non-trivial. Our assumptions now are that $\mathcal{C}'_{G-S}$ is matchable to S in G'_S and vice versa (by the choice of S), so Proposition 8.4.8 yields that G'_S has a 1-factor. This defines a matching of S in G that picks one vertex x_C from every component $C \in \mathcal{C}'_{G-S}$ and leaves the other components of $G - S$ untouched. Adding to this matching a 1-factor of $C - x_C$ for every $C \in \mathcal{C}'_{G-S}$ and a 1-factor of every other component of $G - S$, we obtain the desired 1-factor of G . $\square$

Infinite matching theory may seem rather mature and complete as it stands, but there are still fascinating unsolved problems in the Erdős-Menger spirit concerning related discrete structures, such as partially ordered sets or hypergraphs. We conclude with one about graphs.

Call an infinite graph G *perfect* if every induced subgraph $H \subseteq G$ has a complete subgraph K of order $\chi(H)$, and *strongly perfect* if K can always be chosen so that it meets every colour class of some $\chi(H)$ -colouring of H . (Exercise 75 gives an example of a perfect graph that is not strongly perfect.) Call G *weakly perfect* if the chromatic number of every induced subgraph $H \subseteq G$ is at most the supremum of the orders of its complete subgraphs.

Conjecture. (Aharoni & Korman 1993)

Every weakly perfect graph without infinite independent sets of vertices is strongly perfect.

8.5 Recursive structures

In this section we introduce another tool that is commonly used in infinite graph theory: to define a class of graphs recursively, so as to be able later to prove assertions about these graphs by (transfinite) induction. Rather than attempting a systematic treatment of this technique we give two examples; more can be found in the exercises.

Our first example is very simple: it describes the structure of a tree by recursively pruning away leaves and isolated ends. Let T be any tree, equipped with a root and the corresponding tree-order on its vertices. We recursively label the vertices of T by ordinals, as follows. Given an ordinal α , assume that we have decided for every $\beta < \alpha$ which of the vertices of T to label β , and let T_α be the subgraph of T induced by the vertices that are still unlabelled. Assign label α to every vertex t of T_α whose up-closure $[t]_{T_\alpha} = [t]_T \cap T_\alpha$ in T_α is a chain. The recursion terminates at the first α not used to label any vertex; for this α we put $T_\alpha =: T^*$.

For each α , the vertices labelled α form an up-set in T_α : if $[t]_{T_\alpha}$ is a chain, then so is $[t']_{T_\alpha}$ for every $t' \in [t]_{T_\alpha}$. Every T_α , therefore, is a down-set in T (induction on α) and hence connected. Thus, T_α is a tree, and the set of vertices labelled α induces in T_α a disjoint union of paths.

Let us call T *recursively prunable* if every vertex of T gets labelled in this way, i.e., if $T^* = \emptyset$. We may then be able to prove assertions about T , or about graphs containing T as a normal spanning tree, by dealing in turn with those chains as they get deleted. The following proposition shows that the recursively prunable trees form a natural class also in structural terms: